

Rank and crank are not equidistributed: a counterexample to Conjecture 7292

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Statement being disproved. Conjecture 00000007292 in The Last Math Competition asserts that the rank and crank of integer partitions have the same distribution, and adds a claim about their generating function. We disprove the equidistribution clause using partitions of 4. Here distribution is by the integer value of the statistic, as in the standard definitions below; the problem specifies no modulus or restricted arithmetic progression. Disproving this first clause suffices to disprove the stated conjunction.

Definitions. A partition of n is a finite sequence $\lambda = (\lambda_1, \dots, \lambda_\ell)$ of positive integers in weakly decreasing order, with sum n . For a nonempty partition, its Dyson rank is

$$r(\lambda) = \lambda_1 - \ell.$$

Let $o(\lambda)$ count its parts equal to 1, and let $u(\lambda)$ count its parts strictly greater than $o(\lambda)$. The Andrews–Garvan crank is

$$c(\lambda) = \begin{cases} \lambda_1, & o(\lambda) = 0, \\ u(\lambda) - o(\lambda), & o(\lambda) > 0. \end{cases}$$

These are the standard statistics used by Atkin and Garvan [1, Introduction]. Put

$$N(m, n) = \#\{\lambda \vdash n : r(\lambda) = m\}, \quad M(m, n) = \#\{\lambda \vdash n : c(\lambda) = m\}.$$

Equidistribution would imply $N(m, n) = M(m, n)$ for every integer m and every positive integer n .

Counterexample. The complete list at $n = 4$, with both statistics computed from their definitions, is

λ	ℓ	$o(\lambda)$	$r(\lambda)$	$c(\lambda)$
(4)	1	0	3	4
(3, 1)	2	1	1	0
(2, 2)	2	0	0	2
(2, 1, 1)	3	2	-1	-2
(1, 1, 1, 1)	4	4	-3	-4

For completeness, a partition of 4 has at most four parts. With one part it is (4); with two parts the decreasing positive solutions of $a + b = 4$ are (3, 1) and (2, 2); with three parts only (2, 1, 1) is possible; with four parts all parts equal 1. This proves the list is exhaustive. Hence

$$N(4, 4) = 0 \neq 1 = M(4, 4),$$

contradicting equidistribution. The same discrepancy can be seen without the table: every partition of 4 has largest part at most 4 and at least one part, so its rank is at most 3, whereas (4) has crank 4.

Formal verification and semantic bridge. The self-contained Lean 4.19.0 project imports only `Std`. `IsPartition n parts` is the standard definition: the list is weakly decreasing, every part is positive, and its sum is n . `first_is_largest` verifies that the first list entry used in both statistics is indeed the largest part. Both subtractions take place in `Int`, so negative ranks and cranks are retained.

The theorem `partitionsFour_complete` quantifies over *arbitrary lists of natural numbers*, proves their length is at most four, and classifies each possible length. Together with `partitionsFour_exact` and `partitionsFour_nodup`, this shows that counting the listed objects is precisely counting all standard partitions once, rather than sampling selected partitions.

`RankCrankEquidistribution` states equality of all statistic-fiber counts for every complete duplicate-free enumeration of the partitions of n . Thus its counts are ordinary finite cardinalities, independent of the chosen enumeration. The final theorem `conjecture7292_counterexample` negates this universal assertion by applying it to the verified enumeration at $n = 4$. The supplementary theorem `not_equal_supports` independently formulates the contradiction directly with existential quantifiers over all partitions: crank 4 occurs and rank 4 does not.

The project contains no proof placeholders, native decision procedure, or added axioms. The displayed `#print axioms` audits use only Lean’s standard `propext`, `Classical.choice`, and `Quot.sound`. There is no dependence on a generating-function identity, and the exceptional convention sometimes used for crank generating functions at $n = 1$ is irrelevant to $n = 4$.

References

- [1] A. O. L. Atkin and F. G. Garvan, *Relations between the ranks and cranks of partitions*, Ramanujan J. **7** (2003), 343–366. arXiv:math/0208050.